Course outline (Syllabus of the course) (Template – 1)

Course Code:CS1105 **Name of the Programme: BCA(Honors)** **Name of the Course: Introduction to Version Control** **Semester: II**		
Course credit	**No. of hours per week**	**Total no. of Teaching hours**
02	**0:1:2**	**45**
Course Objectives:		
a) Develop computational thinking and effective problem-solving abilities		
b) Gain hands-on experience with version control, AI tools, and modern technologies.		
c) Encourage collaborative learning and peer-to-peer interaction.		
d) Practice ethical, responsible, and reflective coding habits.		

Syllabus:

Tutorials	**[15 Hours]**
1	Introduction to competitive programming and coding platforms: LeetCode, HackerRank, Codeforces
2	Benefits of practicing on coding platforms; Problem-solving strategies
3	Introduction to Version Control Systems (VCS): Centralized vs. Distributed
4	Installing Git on Windows/Linux/Mac; Setting up user credentials
5	Basic Git operations: git init, git add, git commit, git status, git log
6	Understanding local vs. remote repositories; Git hosting platforms (GitHub, GitLab, Bitbucket)
7	Creating and linking repositories on GitHub; Pushing and pulling changes
8	Forking and cloning repositories; Understanding gitignore
9	Branching in Git: Creating, switching, and deleting branches
10	Merging branches; Identifying and resolving merge conflicts
11	Introduction to GitHub collaboration: Pull requests and code reviews
12	Best practices in reviewing and managing pull requests
13	Advanced Git features: git stash, git rebase vs git merge, git tag, git cherry-pick
14	Introduction to GitHub Co-Pilot; Installing and configuring Co-Pilot in VS Code
15	Ethical use of AI in coding; Best practices: commit messages, branching strategies, clean history

Lab Programs	**[30 Hours]**
1	Create accounts on LeetCode, HackerRank, and Codeforces and solve two problems on each platform.

2	Focus on array and string problems on LeetCode and reflect on solution strategies.
3	Participate in a Codeforces contest and submit a solution analysis report.
5	Install Git and configure user credentials on your system.
6	Initialize a new Git repository and commit a sample file using basic Git commands.
7	View repository status and commit history using git status and git log.
8	Create and link a local repository with GitHub and push commits using git push.
9	Pull updates from a remote repository and manage repository synchronization.
10	Fork and clone a GitHub repository, create a pull request with changes.
11	Create feature branches, switch between them, and merge them using Git.
12	Simulate and resolve a merge conflict in a collaborative project setup.
13	Use git stash to save and apply temporary changes during development.
14	Rebase a feature branch onto the main branch using git rebase
15	Tag specific commits using git tag and cherry-pick selected commits into another branch.

Course outcomes
a) Understand the purpose, structure, and benefits of centralized and distributed version control systems.
b) Develop problem-solving abilities using coding platforms and real-world programming tasks.
c) Demonstrate proficiency in Git and GitHub for collaborative version control and project management.
d) Apply advanced Git techniques and adopt best practices to enhance workflow efficiency and team productivity.

Text books	
1.	Dan Jurafsky and James Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition. Prentice Hall, Second Edition, 2009. ISBN-13: 978-0131873216.
2.	Chris Manning and Hinrich Schütze. Foundations of Statistical Natural Language Processing. MIT Press, Cambridge, MA: May 1999. ISBN-13: 978-0262133609.
3.	Steven Bird, Ewan Klein, and Edward Loper. Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O'Reilly Media, First Edition, June 2009. ISBN-13: 978-0596516499.

Reference books	
1.	Jacob Eisenstein. Introduction to Natural Language Processing. The MIT Press. 2019. ISBN-13: 978-0262042840.

2.	Delip Rao and Brian McMahan. Natural Language Processing with PyTorch: Build Intelligent Language Applications Using Deep Learning. O'Reilly Media. 2019. ISBN-13: 978-1491978238.
3.	Natural Language Processing - Course - Swayam – NPTEL. (https://onlinecourses.nptel.ac.in/noc19_cs56/preview)
4.	Applied Natural Language Processing- Course - Swayam – NPTEL. https://onlinecourses.nptel.ac.in/noc20_cs87/preview)

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Lesson plan (Template –2)

Name of the Programme:	**BCA(Honors)**	
Title of the Course:	**Introduction to Version Control**	
Course code	**CS1105**	
Semester:	**II**	
Names of the Course Instructor(s)	**Dr. K Sailaja Kumar** **Prof. Pranjul Shukla, Prof. Mo. Danish**	
Course credit	**No. of hours per week (0:1:2)**	**Total no. of Teaching hours**
02	**03 Hours**	**45 Hours**

Session	Topic	Pedagogy /activities	Pre-reading/reference
1.	Introduction to coding platforms: LeetCode, HackerRank, Codeforces	Interactive demo and guided exploration of platforms	LeetCode Handbook, HackerRank Docs, Pro Git Ch.1
2.	Introduction to coding platforms	Register on platforms & solve basic problems	LeetCode Practice, HackerRank Tutorials, Codeforces Guide
3.	Introduction to coding platforms	Continue solving problems; compare solutions	LeetCode Practice, Editorials
4.	Problem-solving strategies on coding platforms	Discussion on strategies, time complexity, patterns	LeetCode Blog, HackerRank Practice Guide
5.	Problem-solving practice	Solve curated problems; analyse logic	LeetCode Discuss, Editorials
6.	Problem-solving practice	Timed problem-solving session	HackerRank Interview Prep
7.	Basics of Version Control: Git & VCS	Centralized vs. Distributed; advantages of Git	Pro Git Ch.1, Git-SCM Docs
8.	Git installation & setup	Install Git; configure user credentials	Git Installation Guide
9.	Git initialization	Create repo; first add & commit	Pro Git Ch.2
10.	Basic Git Commands	git add, commit, status, log	Pro Git Ch.2
11.	Basic workflow	Multi-file commits, explore history	Pro Git Ch.2–3

12.	Inspecting differences	Use git diff; modify/delete files	Git-SCM Docs
13.	Remote Repositories	GitHub overview, local vs remote	GitHub Docs
14.	Push local → remote	Create GitHub repo, push commits	GitHub Quickstart
15.	Pulling & syncing	Pull updates; handle basic conflicts	GitHub Docs
16.	Forking and Cloning	Contribution workflow, fork/clone model	GitHub Collaboration Guide
17.	Forking & cloning practice	Fork repo, clone locally, modify	GitHub Classroom
18.	Pull Requests	Submit PR; review peers' PRs	GitHub Docs
19.	Branching in Git	Create, switch, delete branches	Pro Git Ch.3
20.	Branching practice	Create feature branches; edit code	Git Branching Tutorial
21.	Merging branches	Merge features; observe outputs	Pro Git Ch.3
22.	Merge Conflicts	Why conflicts occur; their resolution	Pro Git Ch.3
23.	Conflict simulation	Trigger merge conflicts	Git Docs
24.	Conflict resolution practice	Resolve conflicts step-by-step	Git Merge Tools
25.	Git Stash	Temporary storage of work-in-progress	Pro Git Ch.6
26.	Stash practice	stash, apply, pop, drop	Git Stash Docs
27.	Stash scenario exercises	Use stash within workflows	Pro Git Ch.6
28.	Git Rebase vs Merge	Linear history, reordering commits	Pro Git Ch.6
29.	Rebase practice	Basic + interactive rebase	Git Rebase Docs
30.	Fixing rebase issues	Squashing, editing commits	Git Tools Docs
31.	Git Tags & Cherry-pick	Release tags, isolated commit transfer	Pro Git Ch.6
32.	Tagging practice	Lightweight & annotated tags	Git Tag Docs

33.	Cherry-pick practice	Pick specific commits into branches	Git SCM Docs
34.	GitHub Co-Pilot Introduction	Installation, setup, ethical overview	Co-Pilot Docs
35.	Co-Pilot coding practice	Generate starter code with AI	Co-Pilot Labs
36.	Debugging with Co-Pilot	Check correctness of suggestions	Co-Pilot Labs
37.	Ethical AI Use & Limitations	Responsible coding with AI	AI Ethics Guidelines
38.	Ethical code review	Compare human vs AI code	Co-Pilot Docs
39.	Documentation & reflection	Write reflective notes on AI use	AI Ethics Docs
40.	Best Git Practices	Commit messages, branch strategy	Git Best Practices Guide
41.	Project setup	Create repos; assign roles	GitHub Project Setup
42.	Group work session	Start project implementation	Team Workflow Docs
43.	Final Integration	Complete Git workflow revision	GitHub Docs
44.	Project demonstration	Students present repositories	SEE Practical Guide
45.	Final assessment & viva	Faculty evaluation & submission	Course Notes

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz Graded Component-1 - IBM SkillsBuild Courses: Graded Component-2	 10 marks 10 marks
CIE-2 - Practical Component	25 Marks
CIE-3 - Lab work submission (GitHub) & viva	15 Marks +10 marks

Semester End Examination (SEE): 30%	
Mode of Exam	Practical
Weightage of exam	30 Marks

Course Outcomes- Programme Outcomes Matrix

Sl. No.	Course outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1.	Understand the purpose, structure, and benefits of centralized and distributed version control systems	**3**	**2**	0	**2**	0	0	0	**1**	0	0	0	0
2.	Develop problem-solving abilities using coding platforms and real-world programming tasks	**3**	**3**	**2**	**2**	0	0	0	**2**	0	0	0	0
3.	Demonstrate proficiency in Git and GitHub for collaborative version control and project management	**3**	**2**	0	**3**	**2**	**2**	**2**	**2**	0	0	0	0
4.	Apply advanced Git techniques and adopt best practices to enhance workflow efficiency and team productivity	**3**	**3**	**2**	**3**	**2**	**2**	**3**	**3**	0	**2**	0	0

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE-1 Components		**Rubrics for Assessment**
CIE-1	Quiz Graded Component 1	The quiz will be conducted for 10 marks. Each right answer carries one mark. No negative marking for wrong answers.
	IBM SkillsBuild Courses: Graded Component-2	Certificate of completion for 10 marks. The specified courses on the IBM SkillsBuild platform are to be uploaded to the designated Google Classroom. ***(Refer to Table 1)***

Table-1:

Graded Component-2 (5 Marks):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	**Excellent (10 Marks)**	**Satisfactory (≤7 Marks)**	**Needs Improvement (≤1 Mark)**
Completion of Courses (10 Marks)	Successfully completes all specified courses on the IBM SkillsBuild platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Total Marks		**25**
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Practical Component	The CIE-2 practical examination will be conducted in the laboratory for 25 marks***. (Refer Table-2)*** Writing / Procedure (8 Marks) Execution / Output (12 Marks) Viva-Voce (5 Marks)

Table-2:

CIE-2 (25 Marks): Practical Laboratory Examination

Assess students' practical skills in applying version control concepts using Git and GitHub through hands-on execution and viva-voce.

Criteria	**Excellent**	**Satisfactory**	**Needs Improvement**
Writing the Procedure (8 Marks)	Writes correct Git commands / procedure in proper sequence; shows clear understanding of the task. (8 Marks)	Writes most commands correctly but sequence or explanation has minor errors. (≤5 Marks)	Writes incorrect or incomplete commands; procedure not clear. (≤2 Mark)
Execution and Output (12 Marks)	Executes the task correctly in the system; required Git operations are performed and output is correct. (12 Marks)	Executes most steps correctly; minor execution or output errors. (≤6 Marks)	Unable to execute the task correctly; output is incorrect or missing. (≤3 Mark)
Viva-Voce (5 Marks)	Explains commands, workflow, and output confidently and correctly. (5 Marks)	Gives basic explanation with some hesitation. (≤3 Mark)	Unable to explain commands or workflow. (≤1 Mark)

Total Marks		**25**
CIE-3 Components		**Rubrics for Assessment**
CIE-3	Practical Component	Lab Work Submission (GitHub) & Viva-Voce Total Marks: 25 (15 Marks + 10 Marks) (***Refer Table-3)***

Table-3:

CIE-3: Lab Evaluation (GitHub Submission with Viva-Voce)

Every laboratory exercise shall be evaluated based on work submitted on GitHub, along with a viva-voce conducted for the same submission.

Criteria	**Excellent** (2 Marks)	**Satisfactory** (1 Mark)	**Needs Improvement** (0.5 Marks)
Completion of Lab Work (2 Marks)	All assigned lab exercises are completed and submitted on GitHub.	Most lab exercises are completed; minor omissions.	Several lab exercises are incomplete or not submitted.

Repository Organization & Documentation (2 Marks)	Repository is well-organized with proper folder structure and basic documentation.	Repository structure is acceptable with limited documentation.	Repository is poorly organized with missing documentation.
Commit History & Version Control Practices (2 Marks)	Regular commits with meaningful messages reflecting proper version control practices.	Few commits or generic commit messages.	Very few or single commits; poor version control practices.
Use of Git Commands & Features (2 Marks)	Appropriate use of Git commands.	Limited use of Git commands.	Incorrect or minimal use of Git commands.
Understanding of Submitted Lab Work (2 Marks)	Clearly explains the submitted lab work, commands used, and workflow followed.	Basic explanation with minor gaps.	Unable to explain the submitted work.

Total Marks		**30**
Semester End Examination		**Rubrics for Assessment**
SEE	Practical Component	The SEE practical examination shall be conducted in the laboratory. Evaluation will be based on writing/procedure, execution/output, and viva-voce, with reference to the work submitted on GitHub. ***(Refer to Table 4)*** Writing / Procedure (10 Marks) Execution / Output (15 Marks) Viva-Voce (5 Marks)

Criteria	Excellent	Satisfactory	Needs Improvement
Writing / Procedure (10 Marks)	Writes correct Git commands and procedures in proper sequence; clearly demonstrates understanding of the given task. (10 Marks)	Writes most commands correctly but with minor errors in sequence or explanation. (≤6 Marks)	Writes incorrect, incomplete, or unclear commands; procedure not properly written. (≤3 Marks)
Execution / Output (15 Marks)	Executes the given practical task correctly on the system; all required Git	Executes the task with minor mistakes; output is partially correct. (≤8 Marks)	Unable to execute the task properly; output is incorrect or missing. (≤5 Marks)

	operations are performed, and the output is correct. (10 Marks)		
Viva-Voce (5 Marks)	Explains commands used, GitHub submission, workflow followed, and outputs confidently and accurately. (5 Marks)	Provides basic explanation with hesitation or minor conceptual gaps. (≤3 Marks)	Unable to explain commands, workflow, or output satisfactorily. (≤1 Marks)